AP Precalculus · Unit 1 Checkpoint

Lessons 1.1 to 1.10 — Polynomial and Rational Functions (first ten topics)

Time	50 minutes: Part A (calculator) 30 minutes, then Part B (no calculator) 20 minutes.
Marks	29: Part A 14, Part B 15. Every lettered part is worth 1 mark.
Rules	Show the work that leads to each answer. Calculator in radian mode. Unless told otherwise, give decimal approximations to three places after the decimal point and do not round intermediate values. Express limits in limit notation.
Name / Date	_____ _____

PART A · A GRAPHING CALCULATOR IS REQUIRED · 14 MARKS

1. RATES OF CHANGE · QUADRATIC MODEL **[6]**

A musician released a new song on a streaming service. A streaming service is an online entertainment source that allows users to play music on their computers and mobile devices. Several months later, the musician began using an app (at time $t = 0$) that counts the total number of plays for the song since its release. A "play" is a single stream of the song on the streaming service. The table gives the total number of plays, in thousands, for selected times t months after the musician began using the app. At $t = 0$, the total number of plays was 25 thousand. At $t = 2$, the total number of plays was 30 thousand. At $t = 4$, the total number of plays was 34 thousand.

Months after the musician began using the app	0	2	4
Total number of plays for the song since its release (thousands)	25	30	34

The total number of plays, in thousands, for the song since its release can be modeled by the function D given by $D(t) = at^2 + bt + c$, where $D(t)$ is the total number of plays, in thousands, for the song since its release, and t is the number of months after the musician began using the app.

A. (i) Use the given data to write three equations that can be used to find the values for constants a, b, and c in the expression for $D(t)$. [1]

(ii) Find the values for a, b, and c as decimal approximations. [1]

B. (i) Use the given data to find the average rate of change of the total number of plays for the song, in thousands per month, from $t = 0$ to $t = 4$ months. Express your answer as a decimal approximation. Show the computations that lead to your answer. [1]

(ii) Use the average rate of change found in part B (i) to estimate the total number of plays for the song, in thousands, for $t = 1.5$ months. Show the work that leads to your answer. [1]

(iii) Let A_t represent the estimate of the total number of plays for the song, in thousands, using the average rate of change found in part B (i). For $A_{1.5}$ found in part B (ii), it can be shown that $A_{1.5} < D(1.5)$. Explain why, in general, $A_t < D(t)$ for all t, where $0 < t < 4$. Your explanation should include a reference to the graph of D and its relationship to A_t. [1]

C. The quadratic function model D has exactly one absolute minimum or one absolute maximum. That minimum or maximum can be used to determine a domain restriction for D. Based on the context of the problem, explain how that minimum or maximum can be used to determine a boundary for the domain of D. [1]

Source: College Board, AP Precalculus 2025 Free-Response Questions, Question 2. Reproduced for classroom use.

2. RATES OF CHANGE · CONCAVITY **[3]**

On the initial day of sales ($t = 0$) for a new video game, there were 40 thousand units of the game sold that day. Ninety-one days later ($t = 91$), there were 76 thousand units of the game sold that day.

The number of units of the video game sold on a given day can be modeled by the function G given by $G(t) = a + b\ln(t + 1)$, where $G(t)$ is the number of units sold, in thousands, on day t since the initial day of sales.

B. (i) Use the given data to find the average rate of change of the number of units of the video game sold, in thousands per day, from $t = 0$ to $t = 91$ days. Express your answer as a decimal approximation. Show the computations that lead to your answer. [1]

(ii) Use the average rate of change found in (i) to estimate the number of units of the video game sold, in thousands, on day $t = 50$. Show the work that leads to your answer. [1]

(iii) Let A_t represent the estimate of the number of units of the video game sold, in thousands, using the average rate of change found in (i). For A_{50}, found in (ii), it can be shown that $A_{50} < G(50)$. Explain why, in general, $A_t < G(t)$ for all t, where $0 < t < 91$. [1]

Source: College Board, AP Precalculus 2024 Free-Response Questions, Question 2, part B only (parts A and C are Unit 2 content). Reproduced for classroom use.

3. RATES OF CHANGE · CONCAVITY **[3]**

A person purchased a car at the end of the year 2019 ($t = 0$). The car's value decreases over time. At the end of the year 2020 ($t = 1$), the car was valued at 27.2 thousand dollars. At the end of the year 2025 ($t = 6$), the car was valued at 14.8 thousand dollars.

The value of the car can be modeled by the function V given by $V(t) = ab^t$, where $V(t)$ is the value of the car, in thousands of dollars, at time t, and t is the number of years since the end of 2019.

B. (i) Use the given data to find the average rate of change of the value of the car, in thousands of dollars per year, from $t = 1$ to $t = 6$ years. Express your answer as a decimal approximation. Show the computations that lead to your answer. [1]

(ii) Use the average rate of change found in part B (i) to estimate the value of the car, in thousands of dollars, at $t = 3$ years. Show the work that leads to your answer. [1]

(iii) The average rate of change found in part B (i) can be used to determine the secant line for the graph of V on the interval $1 \leq t \leq 6$. Let $A(t)$ represent the estimate of the value of the car, in thousands of dollars, at time t years, using the secant line. For $A(3)$, found in part B (ii), it can be shown that $A(3) > V(3)$. In general, $A(t) > V(t)$ for all t, where $1 < t < 6$. Use the secant line and the graph of V to explain why this is true. [1]

Source: College Board, AP Precalculus 2026 Free-Response Questions, Question 2, part B only (parts A and C are Unit 2 content). Reproduced for classroom use.

4. POLYNOMIAL ZEROS · END BEHAVIOR **[2]**

The function g is given by $g(x) = -0.167x^3 + x^2 - 1.834$.

B. (i) Find all values of x, as decimal approximations, for which $g(x) = 0$, or indicate that there are no such values. [1]

(ii) Determine the end behavior of g as x increases without bound. Express your answer using the mathematical notation of a limit. [1]

Source: College Board, AP Precalculus 2025 Free-Response Questions, Question 1, part B only (parts A and C are Unit 2 content). Reproduced for classroom use.

End of Part A. Stop and hand in the calculator before Part B.

PART B · NO CALCULATOR · 15 MARKS

Questions 5 to 7 are original AP-style questions written for this checkpoint by Mr Malek Thiab, because the 2024–2026 official free-response papers contain no question on rational functions or on change in tandem.

5. CHANGE IN TANDEM · RATE OF CHANGE **[3]**

A cyclist rides along the Jeddah Corniche. The table gives $d(t)$, the distance in kilometres she has travelled from the start, t hours after she set off.

t (hours)	0	1	2	3	4
$d(t)$ (km)	0	12	22	30	36

A. Based on the table, does $d(t)$ increase or decrease as t increases? Justify your answer. [1]

B. Find the average rate of change of d from $t = 1$ to $t = 3$. Include units. [1]

C. Find the average rate of change of d over each of the four one-hour intervals. Use these values to describe how the rate of change of d is changing, and state the concavity of the graph of d on $0 \le t \le 4$. [1]

6. POLYNOMIAL ZEROS · END BEHAVIOR **[4]**

The polynomial function p has real coefficients and a leading coefficient of 2. Two of its zeros are $x = 3$, with multiplicity 2, and $x = 1 - 2i$.

A. Explain why $x = 1 + 2i$ must also be a zero of p. [1]

B. Assuming p has the least possible degree, state the degree of p and write $p(x)$ as a product of a constant and factors with real coefficients. Do not expand. [1]

C. Describe the end behavior of p as x increases without bound and as x decreases without bound. Use limit notation. [1]

D. Describe the behavior of the graph of p at $x = 3$. Justify your answer using the multiplicity of the zero. [1]

7. RATIONAL FUNCTIONS · ZEROS, ASYMPTOTES, HOLES, END BEHAVIOR **[8]**

The rational function r is given by $r(x) = \dfrac{2x^2 - 2x - 12}{x^2 - 9}$.

A. (i) Write the numerator and the denominator of r in factored form, and state all values of x that are not in the domain of r. [1]

(ii) The graph of r has a hole. Find the coordinates of the hole. Show the work that leads to your answer. [1]

B. (i) Find all zeros of r. [1]

(ii) A student claims that $x = 3$ is a zero of r because $x - 3$ is a factor of the numerator. Is the student correct? Justify your answer. [1]

C. (i) Write the equation of the vertical asymptote of the graph of r. [1]

(ii) Determine $\lim\limits_{x \to -3^-} r(x)$ and $\lim\limits_{x \to -3^+} r(x)$. Use limit notation. [1]

D. (i) Determine $\lim\limits_{x \to \infty} r(x)$. Use limit notation. [1]

(ii) Write the equation of the horizontal asymptote and explain, using the degrees and leading coefficients, why it is the same as x decreases without bound. [1]

End of checkpoint.